

SOHEN PATEL

Toronto, ON, Canada | (416) 666-8954 | sohenpatel.work@gmail.com | linkedin.com/in/sohen-patel | github.com/sohenpatel22 | Portfolio

TECHNICAL SKILLS

Languages & Data: Python, SQL (MySQL, PostgreSQL), Pandas, NumPy, Statsmodels, Matplotlib, Seaborn, Plotly, Power BI
ML & Deep Learning: scikit-learn, PyTorch, Computer Vision (MediaPipe, OpenCV), Model Evaluation (F1, NDCG, RAGAS, DeepEval)
LLMs: LangChain, LangGraph, Langfuse, RAG (hybrid search, Qdrant), DSPy, Claude/OpenAI APIs, Pydantic, Streamlit, Hugging Face
MLOps: Git, GitHub Actions, Docker, FastAPI, MLflow, DVC, pytest, Prometheus/Grafana, ONNX, AWS (S3, EC2, SageMaker, Bedrock)

EXPERIENCE

AI Research Intern

Jul. 2026 - Sep. 2026

Ontario Health

Toronto, ON, Canada

- Built an enterprise Ontario Health News Agent in Microsoft Copilot Studio end-to-end, deployed to 5,000+ employees org-wide
- Created a 259-item human-verified gold-standard dataset for 3 healthcare tasks; built an end-to-end LLM evaluation framework for 5 models (GPT-5, Claude, Gemini, MS Copilot, Perplexity) across 13+ metrics using RAGAS, DeepEval, and LLM-as-a-Judge
- Conducted systematic prompt-engineering experiments across 4 strategies, finding Chain-of-Thought prompting on a 5-year dataset increased agent F1 score from 0.25 to 0.42 (+68%) and reduced hallucination rate by 51% (78% to 27%); logged runs to Langfuse.

AI Engineering Project Team Lead

May 2026 - Jun. 2026

Deloitte (Sponsored Practicum)

Toronto, ON, Canada

- Built a 6-agent system that turns a client brief into consulting deliverables (WBS, RACI, RAID, weekly status report) with a human-in-the-loop revision loop, plus Deloitte template sync for DOCX/XLSX exports, backed by 129 automated tests
- Added an LLM routing layer with fallback across 4 backends (Claude, GPT, Gemini, DeepSeek) and prompt-cache warm-up to cut repeated token cost on agent calls
- Built a 13-rule Quality Gates engine (graph-based WBS cycle detection, RAID/RACI cross-artifact validation)

Graduate Engineer Trainee

Aug. 2024 - Aug. 2025

ArcelorMittal Nippon Steel India (AM/NS)

Surat, India

- Used SAP datasets and project logs for real-time monitoring of compliance metrics and operational workflows, generating actionable insights that supported executive-level decision-making across 12 cross-functional departments
- Designed interactive Power BI dashboards to track project KPIs, resource allocation, and progress trends for data-driven planning across a large-scale plant expansion project from 7 to 14 million metric tons per annum (MMTPA)
- Identified workflow bottlenecks and execution gaps, contributing to a 16.7% reduction in project timelines

PROJECTS

ASL Fingerspelling Recognition | Python, PyTorch, MediaPipe, OpenCV, NumPy, Gradio

[GitHub](#)

- Trained 27.8M parameter Conformer-Transformer on 115K+ sequences with 84-dimensional features with wrist normalization
- Optimized beam search via grid search achieving test CER of 0.44 (56% character accuracy) across a 62-token vocabulary
- Deployed inference pipeline with LLM post-processing and text-to-speech synthesis via Gradio and Hugging Face

Two-Stage Recommendation System: Retrieval and LambdaRank Ranking | PyTorch, InfoNCE, LambdaRank

[GitHub](#)

- Built a two-stage recommendation system with a Two-Tower retrieval model and candidate-aware MF reranker
- Implemented contrastive learning with hard negative sampling and optimized ranking using LambdaRank loss
- Achieved 46% improvement in NDCG@10 and +18 percentage points in Hit Rate@10 over the MF baseline

Probabilistic Fraud Detection using Gaussian Mixture Models | scikit-learn, Pandas, NumPy

[GitHub](#)

- Designed unsupervised anomaly detection pipeline for credit card fraud using dual GMMs on imbalanced dataset (~0.17% fraud)
- Formulated fraud detection as a log-likelihood ratio problem with F1-optimized threshold selection using O(n log n) search
- Achieved ROC-AUC of 0.969, PR-AUC of 0.697 (primary metric), and F1-score of 0.727 on a held-out test set

Financial Time Series Forecasting - MSFT Returns | Statsmodels, PyTorch, Scikit-learn

[GitHub](#)

- Developed forecasting pipeline for 5-day MSFT log returns using 12+ models including ARIMA, LASSO, Elastic Net, and LSTM
- Built multi-asset feature set with technical (RSI, MACD), macroeconomic, rolling beta, and lag signals for predictive modeling
- Identified ARIMA as the best-performing model with the lowest MSE of 0.000875 and 62.2% directional accuracy on test data

EDUCATION

University of Toronto

Sep. 2025 - Expected Jan. 2027

Master of Engineering | Emphasis: Data Analytics and Machine Learning | GPA: 3.81/4.0

Toronto, ON

- Coursework: Deep Learning, Foundations of Data Analytics and Machine Learning, Data Science Methods and Statistical Learning, AI in Finance, Introduction to ML | MEng Research Project: Building Multi-Agent AI Systems for Legal Applications

L.D. College of Engineering, Gujarat Technological University

Jul. 2021 - Jul. 2024

Bachelor of Engineering in Mechanical Engineering | CGPA: 8.40/10

Ahmedabad, India

AWARDS, ACHIEVEMENTS & ACTIVITIES

- Awarded Aga Khan Foundation International Scholarship, fully covering graduate study costs at the University of Toronto
- Teaching Assistant for MIE354 Business Process Engineering and MIE1628 Cloud-Based Data Analytics at University of Toronto
- Filed 2 provisional patents for ISRO research project on deployable space telescope systems (IN:202421088072, IN:202421087180)
- Achieved All-India Rank (AIR) 3465 in GATE XE 2023, a national-level engineering examination